

Chandru Ganeshan

AI/ML Engineer - GenAI, RAG Systems, Healthcare ML — India

GitHub: [itzmechandruganeshan](#) | LinkedIn: [chandru-g24](#) | Portfolio

PROFESSIONAL SUMMARY

AI/ML Engineer with 2 years of experience designing and deploying production-grade GenAI and machine learning systems. Specialized in LLM-powered applications, semantic search, RAG pipelines, and healthcare ML solutions. Experienced in building agent-based architectures, large-scale embedding systems, and clinical ML tools used in real-world environments.

PROFESSIONAL EXPERIENCE

Cloudside Technologies Pvt Ltd

Jan 2025 – Present

AI / ML Engineer

- **Tpay - AI-Powered Financial Chatbot Platform (POC)**

- * Architected an agent-orchestrated financial chatbot enabling natural-language queries over structured financial data without SQL knowledge.
- * Designed routing logic across FAQ retrieval, SQL generation, analytics, and forecasting agents within a single conversational flow.
- * Implemented multilingual semantic search (English and Arabic) using FAISS vector stores with automatic language detection.
- * Built natural language-to-SQL pipelines executing dynamic queries on BigQuery tables with currency normalization and formatted outputs.
- * Enabled multi-company data isolation and context-aware querying across 15+ financial categories.
- * Technologies: Python, FastAPI, LangChain, FAISS, Google Vertex AI, BigQuery

- **Marketing & Sales Agent - In3 (POC)**

- * Designed a modular multi-agent system automating lead generation, enrichment, and multi-channel outreach workflows.
- * Built and iterated 5+ specialized agents for marketing, sales, and campaign execution.
- * Integrated CRM APIs and deployed containerized services with a UI for campaign management.
- * Technologies: Python, Gemini LLM, Vertex AI, Docker, Streamlit, Pandas, Google ADK, Veo 3, Google's Nano Banana

ISPG Technologies Pvt Ltd

May 2024 – Dec 2024

GenAI / ML Engineer Intern

- **Production Semantic Search Engine - Mopare (Automotive E-commerce)**

- * Independently architected and deployed a production-grade RAG-based semantic search system for automotive parts discovery.
- * Indexed millions of products with vehicle fitment compatibility data using transformer embeddings and parallel vector search.
- * Achieved sub-500ms query latency through concurrent processing and optimized routing logic.
- * Deployed containerized FastAPI services on Kubernetes with Jenkins-based CI/CD pipelines.
- * Technologies: Python, FastAPI, Sentence Transformers, Docker, AWS Bedrock

- **Production Semantic Search Engine - PNG Jewellery**

- * Built an end-to-end semantic product search engine supporting complex natural language queries with pricing constraints.
- * Implemented embedding-based similarity matching, NLP-driven price extraction, and relevancy validation logic.
- * Processed 10,000+ product category hierarchies with precomputed BERT embeddings.
- * Achieved 100% precision by eliminating false-positive results and reduced latency by ~40%.
- * Technologies: Python, FastAPI, Sentence Transformers (BERT), spaCy, Docker

Microbiological Laboratory Pvt Ltd

Jan 2024 – April 2024

Data Scientist Intern

- **PyMLRS - ML-Based Pathogen Detection System**

- * Architected and developed an end-to-end ML system to automate meningoencephalitis pathogen detection from RT-PCR data.
- * Engineered signal-processing pipelines for HRM curves and amplification cycles with feature extraction and ML classification.
- * Supported detection of 8 clinical pathogens with automated, clinical-grade PDF diagnostic reports.
- * Reduced diagnostic interpretation time from hours to minutes, enabling labs to process 100+ samples per day.
- * Published as a production-ready PyPI package for healthcare workflows.
- * Technologies: Python, SciPy, NumPy, Pandas, Scikit-learn, Matplotlib, Plotly

PROJECTS

PyMLRS - ML-Based Pathogen Detection System

- Published PyPI package: <https://pypi.org/project/PyMLRS/>

Rextractor

- Open-source tool for parsing proprietary RT-PCR.rex files into structured datasets.
- GitHub: <https://github.com/itzmechandruganeshan/RETRACTOR>

TECHNICAL SKILLS

Programming & Backend: Python, FastAPI, Flask, REST APIs, Async Processing

GenAI & NLP: LLMs (Gemini, Mistral), LangChain, LangGraph, Google ADK, RAG Pipelines, FAISS, Semantic Search, Vector Embeddings

Agentic AI & Evaluation: Agentic Frameworks, Ragas

Machine Learning & Frameworks: TensorFlow, Scikit-Learn, Feature Engineering, Classical ML Models, Embeddings, Model Evaluation

Cloud & Deployment: Google Cloud Platform (Vertex AI, BigQuery), Docker, Kubernetes (working knowledge)

Data & Visualization: Pandas, NumPy, SciPy, Matplotlib, Plotly

PROOF OF WORK

GitHub: <https://github.com/itzmechandruganeshan>

PyPI: PyMLRS

Technical Posts: LinkedIn (PyMLRS, RAG Chunking Strategies)